

# Matthew Dong

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## EXPERIENCE

### NJStat

Bridgewater, NJ

#### Artificial Intelligence Intern

Jan 2025 – Present

- Updated internal **natural language processing** AI software using the **Transformers** library in **Python** to improve employee monitoring by 20%.
- Leveraged **zero-shot** and **chain-of-thought prompt engineering** techniques in lieu of fine-tuning to overcome limited computing resources and drive project success.

### Colgate-Palmolive

Piscataway, NJ

#### Application and Software Development Intern (E-Commerce DevOps)

Jun 2022 – Dec 2024

- Developed **Python** scripts that utilize **cloud functions** and **API gateways** to parse **JSON** alerts for various applications and send them to Google Chat spaces using **webhooks**, clearing cluttered email inboxes.
- Automated code updates by implementing robust **CI/CD** workflows using **GitHub Actions** while also maximizing reusability—deploying **Python** scripts to **Google Cloud Platform** as **cloud functions** that interface with **API gateways**.
- Delivered presentations to non-technical audiences, using non-technical jargon as well as **Splunk** dashboards to effectively demonstrate the importance of our work to management.

### Sanofi

Bridgewater, NJ

#### Data Analyst Intern

Dec 2020 – Feb 2021

- Utilized **R** in order to analyze COVID-19 data, while keeping statistical principles in mind.
- Modified and analyzed movie data utilizing machine learning topics: **PCA**, **bootstrapping**, and **decision trees**.

## PERSONAL PROJECTS

(More projects available on GitHub: [github.com/Matt-J-Dong](https://github.com/Matt-J-Dong))

### VentureAI (Python: PyTorch)

- Developed an automated travel agent that utilizes fine-tuned **open-source LLMs** that provides personalized travel recommendations tailored to each user's preferences by integrating the LLM with APIs for dynamic **retrieval augmented generation** (Dynamic RAG)
- Addressed limitations of state of the art models such as hallucination and implemented key technical innovations such as **LoRA** and **quantization**.

### Multicore Chess (C++: OpenMP)

- Greatly improved the speed of move calculations, achieving up to **12x** speedup in comparison to the sequential version.
- Optimized the evaluation function and **parallel alpha-beta pruning** algorithms despite limited CPUs and no access to GPUs.

### Top-Towns-To-Take-Over-Tech (Spark Scala, Java MapReduce, Shell)

- Built an analytic with a group to find out which American cities are the best for tech careers.
- We cleaned, profiled, and performed ETL on our data before joining all the datasets together, and displayed our results through a Tableau visualization and a Hive warehouse.

### Quizverse (MERN Stack)

- Designed and implemented a gamified, spaced-repetition flashcard app that enhances learning retention by scheduling daily review sessions, tracking study streaks, and awarding points for consistent progress.
- Utilized the flexibility of **NoSQL** databases to store user information with JSON and **MongoDB**.

## EDUCATION

### New York University

New York, NY

#### Master of Science: Computer Science

GPA: 3.95/4.00 — Sep 2024 – May 2025

- Coursework: DevOps and Agile Methodologies, Big Data Application Development, Database Systems, Programming Languages, Multicore Processors, Deep Learning

### New York University

New York, NY

#### Bachelor of Arts: Computer Science; Bachelor of Arts: Data Science

GPA: 3.93/4.00 — Sep 2021 – May 2024

- Coursework: Fundamentals of Machine Learning, Advanced Topics: Deep Learning, Responsible Data Science, Basic Algorithms, Operating Systems, Big Data for Analytics Applications, Linear Algebra, Prob and Stat

## SKILLS

**Languages:** Python, R, Markdown, C++, Spark Scala, SQL, R, Bash, Java, x86-64 Assembly

**Libraries:** PyTorch, Transformers, Wandb, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, OpenMP, MPI

**Technologies:** GitHub, GitHub Actions, Hadoop, Kubernetes (K8s), Docker, GCP, Terraform, Jira, Confluence, Regex, HTML, Splunk, Linux, MongoDB

## ORGANIZATIONS/LEADERSHIP

### Intern Social Committee

Piscataway, NJ

#### Head of Internal Operations

Jun 2022 – Dec 2024

- Wrote meeting agendas, coordinated and led weekly meetings, and delegated tasks to achieve the end goal of the committee was to design fun programs for the interns of Colgate to enjoy, received highly positive feedback from the attendees of events.